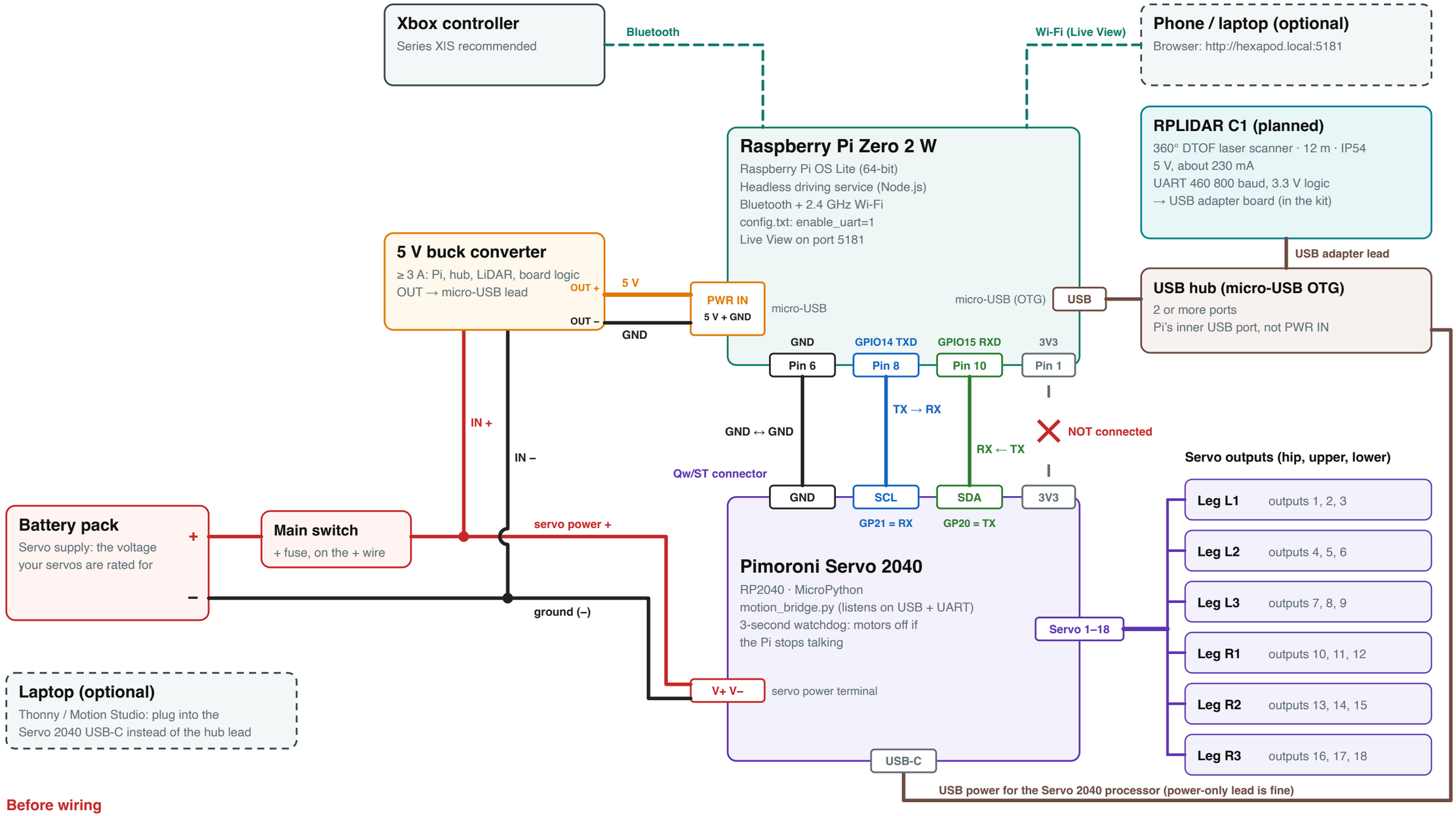
Hexapod — Raspberry Pi Zero 2 W wiring schematic

Xbox controller (Bluetooth) → Pi Zero 2 W → UART on the Servo 2040 SDA/SCL pins → 18 servos · RPLIDAR C1 on USB (planned)



Before wiring

1. VERIFY on the Servo 2040 pinout that SDA = GP20 and SCL = GP21 (UART1 TX/RX).
2. Cross the wires: Pi TX → Servo RX, Pi RX ← Servo TX. Join GND. Do NOT join 3V3.
3. Both boards use 3.3 V logic: no level shifter. Keep UART wires under 20–30 cm.
4. Power the Pi from its own 5 V regulator (5 V and GND), never from the servo rail directly.
5. Common ground: battery –, buck converter, Pi and Servo 2040 GND are all joined.
6. Switch the servo power off before connecting or disconnecting anything.
7. RPLIDAR C1: connect through its USB adapter and the hub. The Pi's GPIO UART is used by the Servo 2040 and its other UART by Bluetooth.

Hexapod Pi Zero 2 W

Wiring schematic · Rev C

5 November 2026

Built and tested; LiDAR planned

LiDAR: check pinout on its datasheet

Legend Battery / servo power + · Ground (–) · 5 V for the Pi · UART: Pi → Servo 2040 · UART: Servo 2040 → Pi · Wireless · USB (hub, LiDAR, board power)